Witchayaphat Pakkhemaya

Mechatronics Engineering Student at KMITL — Mechanical Design, CAD/CAM & Industrial Automation

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EXPERIENCE & PROJECTS

Delta Academy — Zero-Downtime Changeover Retrofit — Sep 2026

Top 3 of 12 teams

- Co-designed a retrofit system that lets an existing machine switch product recipes without shutdown or manual part removal, reducing changeover downtime and improving machine availability.
- Integrated the full Delta automation stack: PLC, HMI, servo and inverter, applied after completing Delta's industrial training program.
- Worked with the team on the pitch: created an operation video of the system so the judges could see the concept working; selected as 1 of 3 finalists from 12 teams.

3D Printing Service — Founder/Operator — Sep 2026 – Present

- Reverse-engineered and 3D-modeled replacement parts for customers who had no drawings or CAD files, then printed functional parts (e.g. drone components).
- Handled material selection based on each part's strength, heat and use requirements.
- Built a cost & profit calculator website for quoting, plus AI auto-reply chat, auto-PR posting and a purchasing workflow.

CNC Machining — CAD/CAM — 2024 – Present

- Programmed and ran CNC turning and milling jobs with Fusion 360 CAD/CAM: G-code posting, machine setup and part inspection.
- 75-hr CAD-CAM/CNC certification (Phra Nakhon Polytechnic College, 2023).

ABU Robocon 2025 — *5th of 24 teams* — Nov 2025 – May 2026

- Planned the system architecture before coding; switched from a ROS2 autonomy plan to a behavior tree to meet the deadline while keeping the robots fully autonomous.
- Troubleshot real engineering problems such as gear backlash and PID tuning on a mecanum drive, and applied initial fixes.
- Kept the project moving despite inconsistent team attendance by re-planning tasks around the members who were available.

NTU Hackathon — Stroke Rehabilitation Game — Apr 2026

- Generated ideas for a game-based tool that makes post-therapy exercise engaging for stroke patients, based on patient-needs research provided by the organizers.
- Learned Unity during the hackathon, the same engine used for industrial digital-twin simulation.

Industrial Internship — Summer 2025 (after Year 1)

- Assisted engineers on-site: PLC panel wiring and mechanical assembly.

Other projects: Line-following robot with path memory (ESP32) · Self-balancing robot (PID control) · Siemens PLC labs (CNC & tank-level control)

SKILLS

Design & Manufacturing: Fusion 360 (CAD/CAM), 3D modeling, reverse engineering, CNC turning & milling, G-code, 3D printing, material selection
Industrial Automation: Delta PLC, HMI, servo & inverter; Siemens PLC (basic); ladder logic; retrofit for downtime reduction
Control & Troubleshooting: PID control, sensor integration (encoder, gyro, IR, ultrasonic), gear backlash troubleshooting
Simulation: Unity (digital-twin exposure), Gazebo (beginner)
Other: Python, C, Arduino, ESP32, ROS2

LEADERSHIP & TEAMWORK

- President of H-Room (cross-year reunion, 4 years)
- Presented control-systems project at KMITL Main Auditorium
- Taught control concepts to Grade 10–12 students via the self-balancing robot
- Organised peer tutoring groups

EDUCATION

B.Eng. Mechatronics Engineering

King Mongkut's Institute of Technology Ladkrabang (KMITL) · GPA 3.38 · 2024 – Present

LANGUAGES

Thai (Native) · English (B1 — KMITL TEP)